Nutrition Assistant Project
Phase 3: Project Design

Introduction

The Project Design phase transforms the requirements into a detailed system blueprint. It defines the architecture, database structure, user interface, and technology stack that will be used to build the Nutrition Assistant application. A well-planned design improves scalability, maintainability, and user experience.

System Architecture

The application follows a three-tier architecture consisting of the Presentation Layer (Frontend), Application Layer (Backend), and Data Layer (Database). The frontend communicates with the backend using RESTful APIs over HTTPS. The backend processes requests, validates users, performs calorie calculations, and interacts with the database. JSON is used as the data exchange format between client and server.

Typical API endpoints include user authentication, food search, meal logging, dashboard retrieval, profile management, and nutrition recommendations. The modular API structure makes future enhancements easier.

Database Schema

Users Table: Stores user ID, name, email, encrypted password, age, height, weight, BMI, calorie goal, and dietary preferences.

Logs Table: Stores meal entries including food name, calories, carbohydrates, proteins, fats, quantity, meal type, date, and user ID.

Additional optional tables include Food Database, Goals, Notifications, and Water Intake.

UI/UX Design

The user interface is designed with simplicity and accessibility in mind. The home screen displays calorie summaries, nutrition charts, and quick-access buttons for logging meals. Navigation is intuitive with a dashboard, search page, meal history, and profile settings.

Wireframes and mockups are prepared before implementation to validate user flow. Design tools such as Figma or Adobe XD can be used for creating interactive prototypes.

Design Guidelines

Primary Colors: Green and White to represent health and freshness. Typography: Clean fonts such as Poppins or Roboto. Responsive layouts for desktop, tablet, and mobile devices. Consistent icons and reusable UI components. Simple navigation with minimal user effort.

Technology Stack

Frontend: React.js / HTML, CSS, JavaScript.
Backend: Node.js with Express or Spring Boot.
Database: MySQL or MongoDB.
Authentication: JWT with encrypted passwords.
Hosting: Vercel, Netlify, or AWS.

Conclusion

The Project Design phase provides a complete blueprint for implementation. It ensures all stakeholders understand the application structure, database relationships, API communication, and interface design before development begins, reducing errors and improving development efficiency.